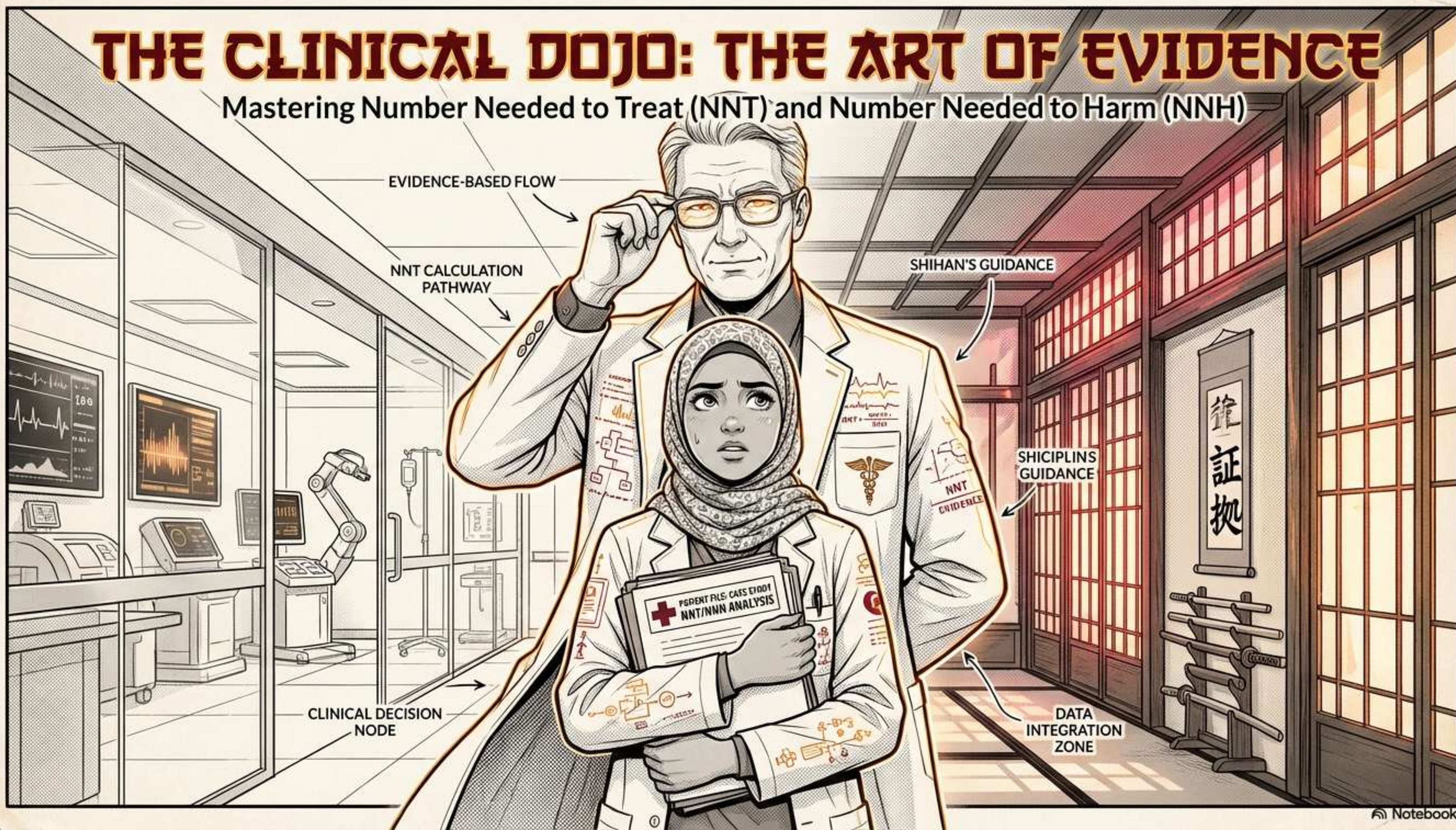


THE CLINICAL DOJO: THE ART OF EVIDENCE

Mastering Number Needed to Treat (NNT) and Number Needed to Harm (NNH)



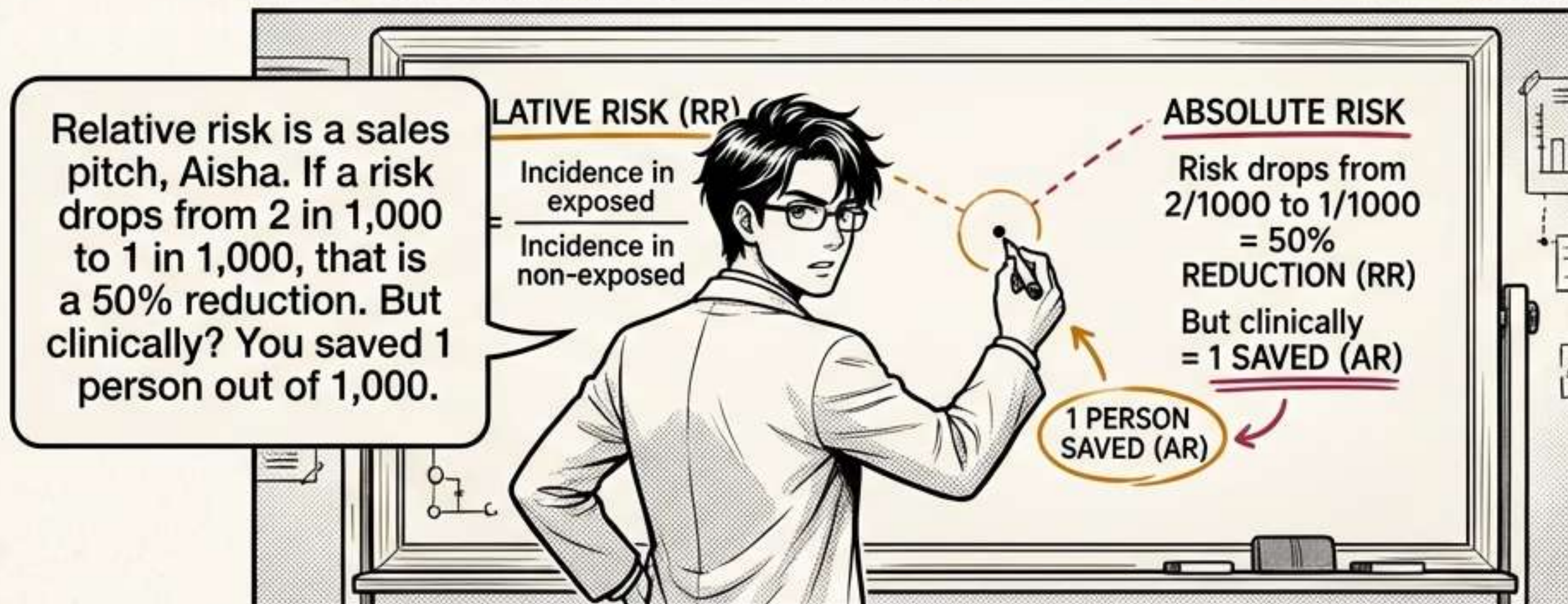
Dr. Aisha, this says it cuts my risk in half! But the side effects sound deadly. What should I do?

**WONDER DRUG
REDUCES STROKE
RISK BY 50%!**

**DRUG CAUSES
CANCER!**

50% reduction... that sounds incredible. But is it the whole story?

**THE INCITING INCIDENT:
THE RELATIVE RISK TRAP**



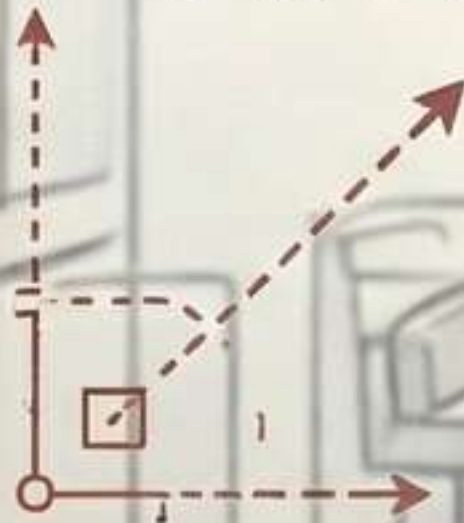
NUMBER NEEDED TO TREAT (NNT)

The number of patients you need to treat to prevent one additional bad outcome.

$$NNT = \frac{1}{ARR}$$

(Absolute Risk Reduction)

It is the reciprocal of the absolute difference. It turns abstract percentages into actual people.



CASE STUDY: ANOTHER ABAN TRIAL

Control Group:
Risk of Stroke
= 4% (0.04)

Absolute Risk Reduction
(ARR) = 4% - 2%
= 2% (0.02)

Calculation:
 $1 / 0.02 = 50$

Treatment Group:
Risk of Stroke
= 2% (0.02)

NNT = 50

So, we treat
50 people for
2 years just to
save 1 from
a stroke?



VISUALIZING NNT = 50



For an NNT of 50: 49 patients take the medication and experience no difference in this specific outcome. **Only 1 patient** benefits.

THE GOLDEN RULES OF NNT

Ideal NNT
2-5



Minor Headache
(Analgesics)

Acceptable NNT
40-100

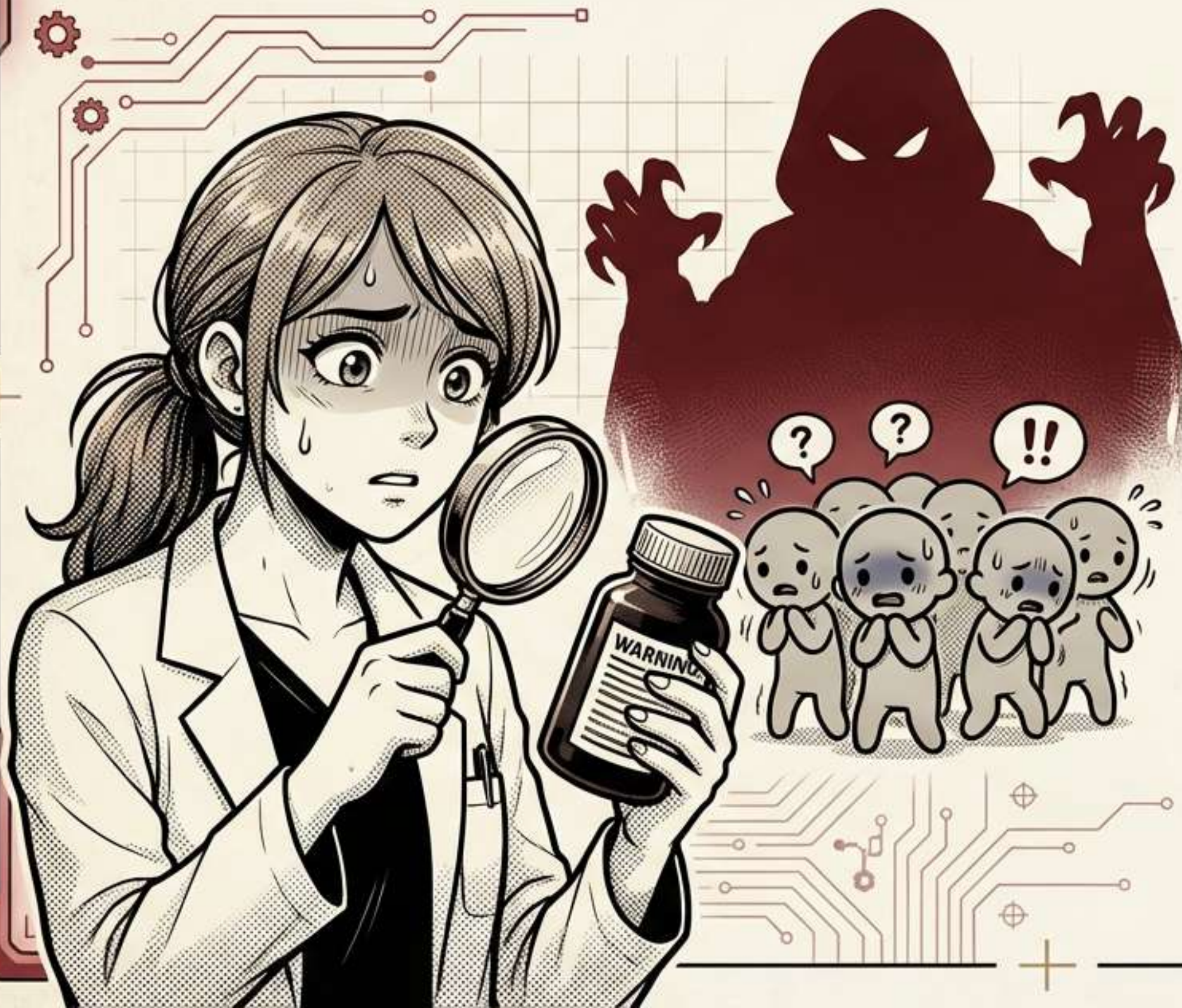


Stroke/Death
(Statins)

An NNT of 50 is acceptable to prevent a stroke. It would be unacceptable for curing a headache. **Context is everything.**

Dr. Kenji

THE SHADOW: NUMBER NEEDED TO HARM (NNH)



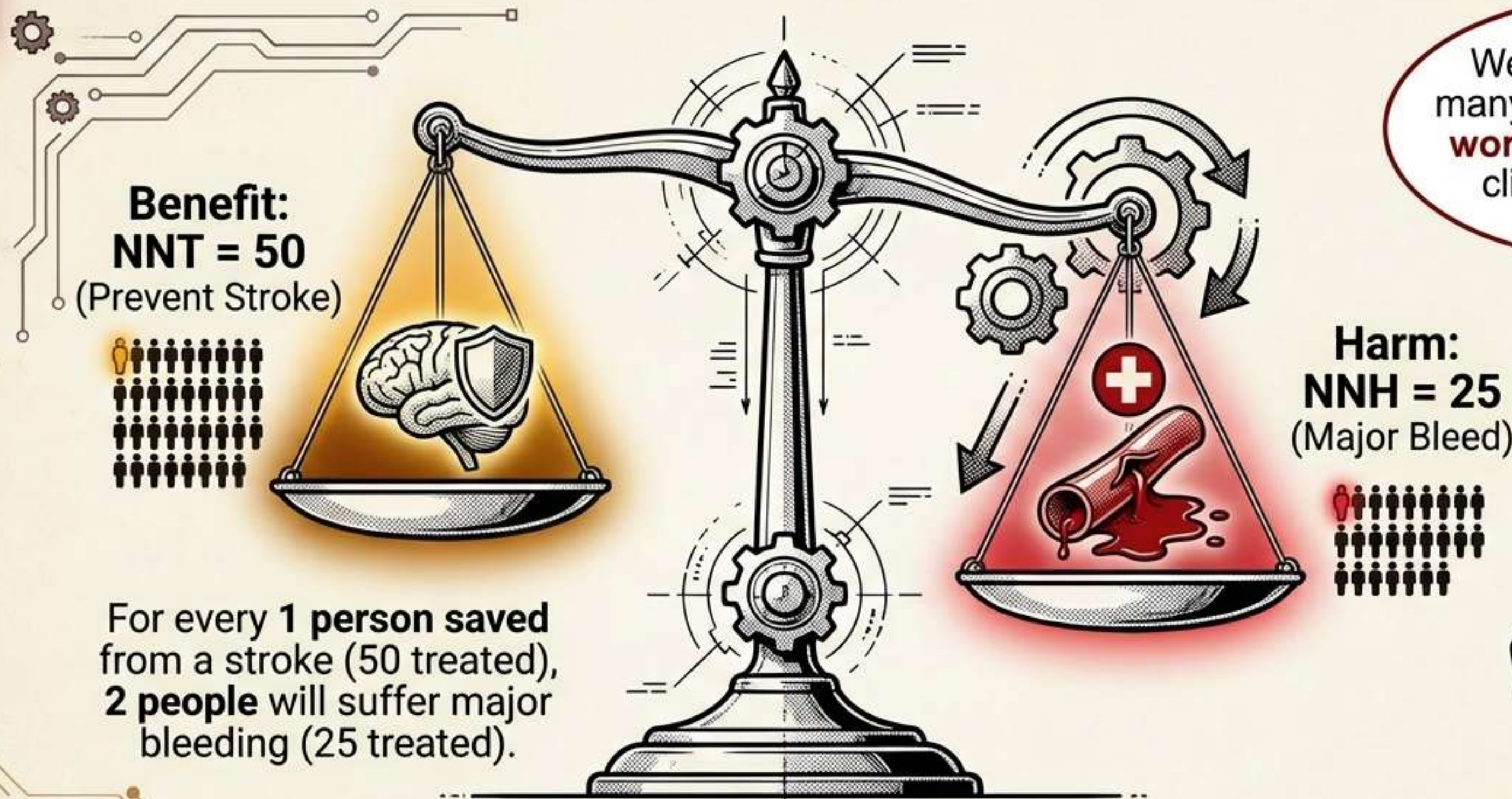
Definition: The number of patients treated for one to suffer an adverse event.

Formula:
$$NNH = 1 / ARI \text{ (Absolute Risk Increase)}$$

Example (Anotheraban Bleeding Risk):
 $ARI = 4\%.$ $NNH = 1 / 0.04 = 25.$

NNH = 25

LIKELIHOOD TO BE HELPED OR HARMED (LHH)



We harm twice as many as we save. Is it **worth it**? That is the clinical question.



Dr. Kenji

ACT IV: THE FINAL EXAM

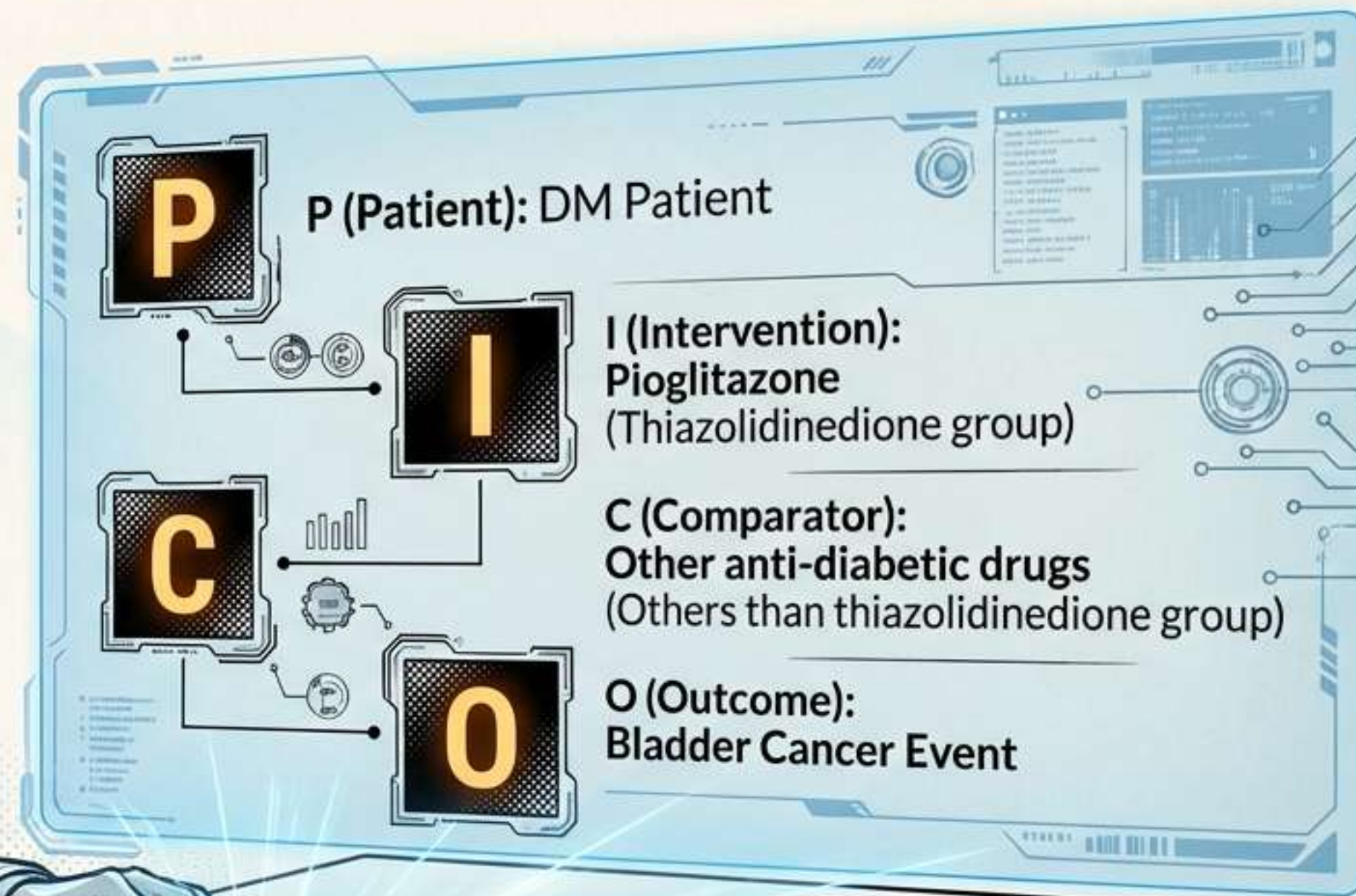
QUEST RECEIVED

Doctor, am I
getting cancer
from this pill?

CASE FILE: PIOGLITAZONE & BLADDER CANCER

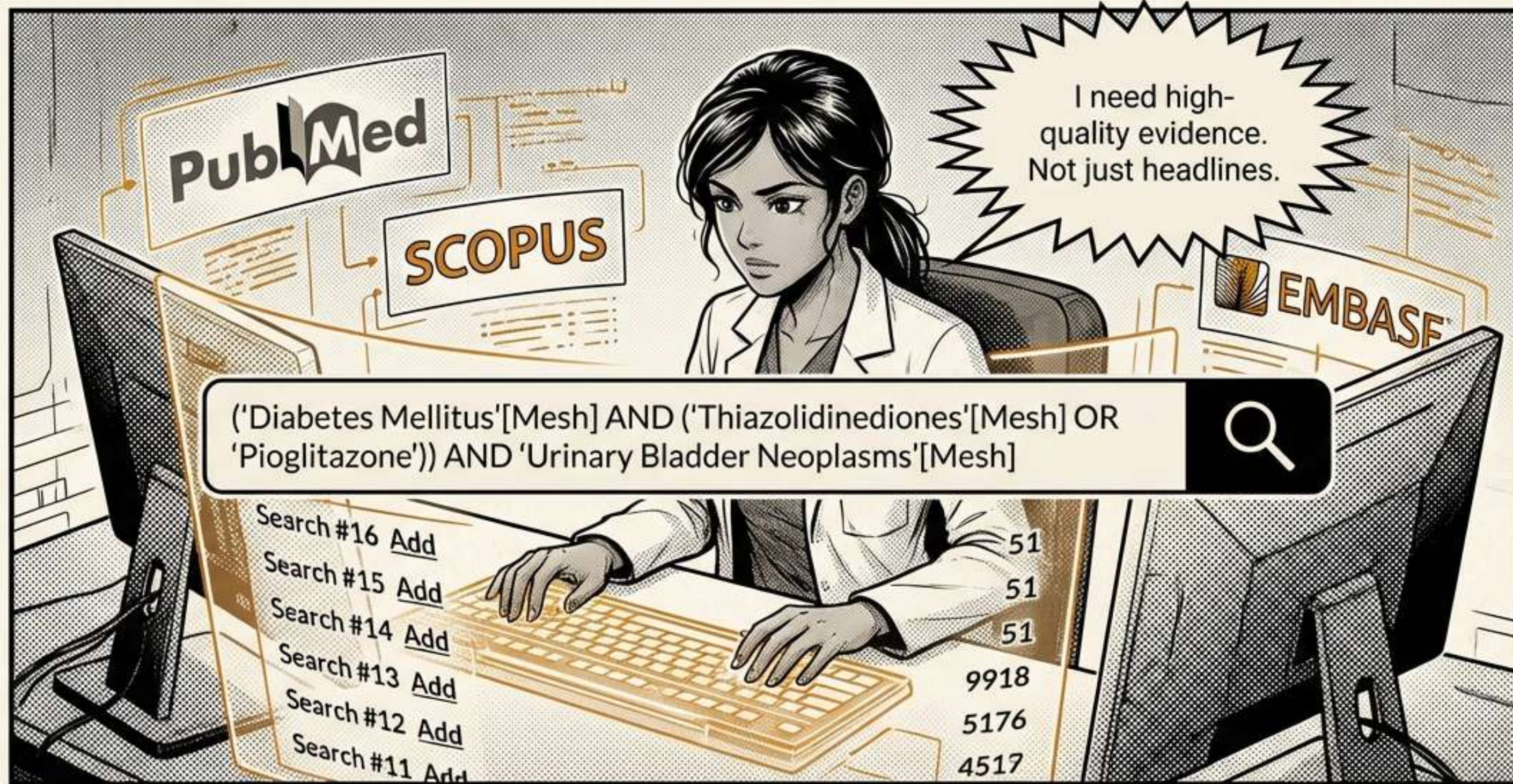
- 65-year-old Male, Diabetes Mellitus.
- Failed Glipizide & Metformin.
- Currently on Pioglitazone (HbA1c = 5).
- **CONCERN:** Read newspaper linking drug to Bladder Cancer.

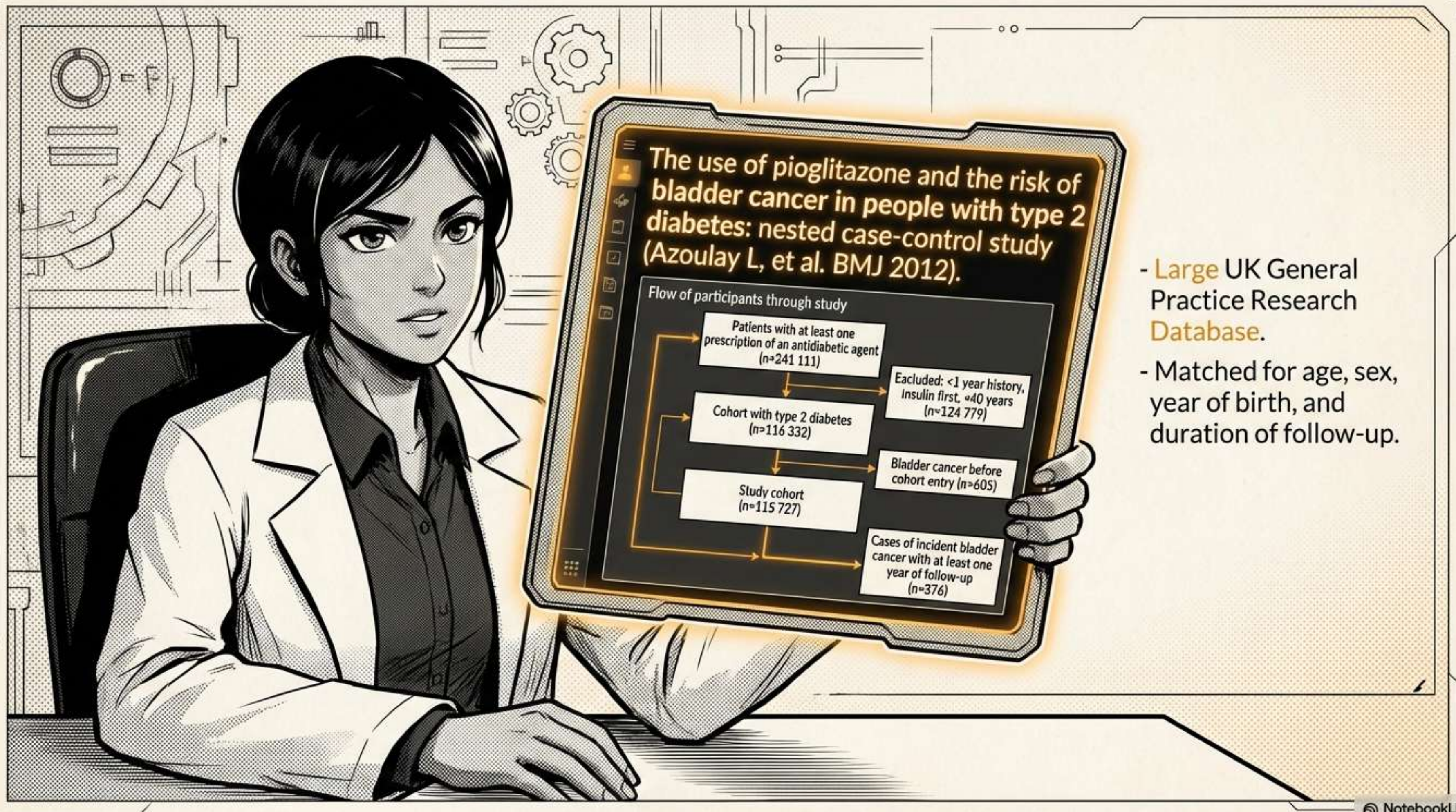
TRANSFORM CLINICAL QUESTION TO **PICO**



Now, to structure the clinical question using PICO for the search...

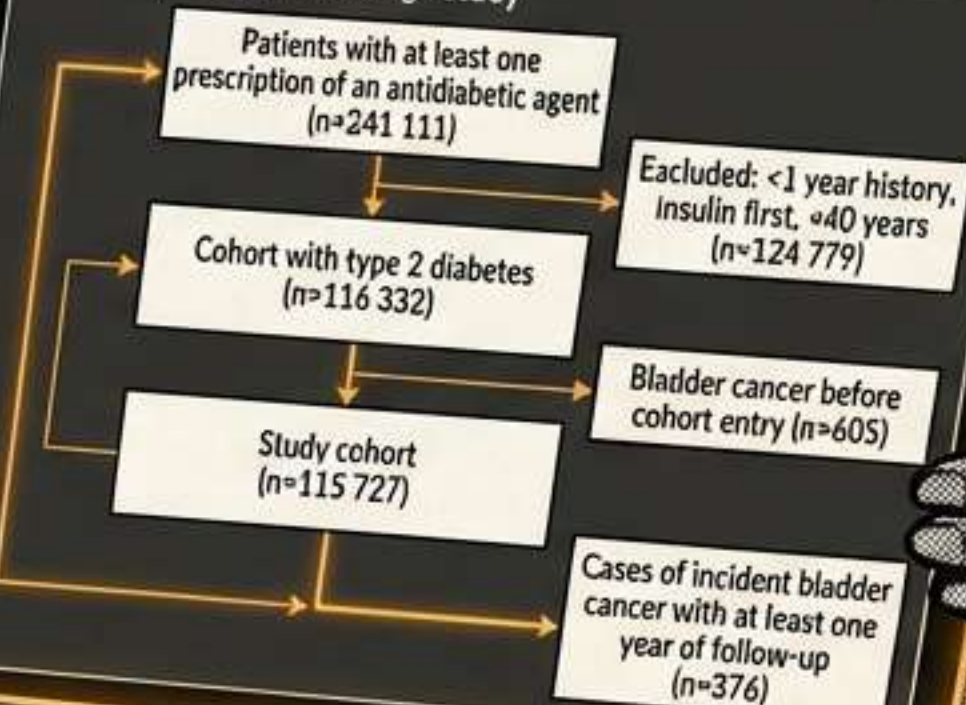
EXECUTING THE SEARCH STRATEGY





The use of pioglitazone and the risk of bladder cancer in people with type 2 diabetes: nested case-control study (Azoulay L, et al. BMJ 2012).

Flow of participants through study



- Large UK General Practice Research Database.
- Matched for age, sex, year of birth, and duration of follow-up.

CRITICAL APPRAISAL: IS IT VALID?

- ✓ 1. Selection Bias? **MINIMIZED** (Matched 1:20).
- ✓ 2. Confounding? **ADJUSTED** (Smoking, Alcohol, Obesity, HbA1c).
- ✓ 3. Exposure Measurement? **RELIABLE** (Prescription records).

Good. The foundation is solid. Now, look at the numbers.

Table 1 Baseline characteristics of bladder cancer cases and controls. Values are number (percentages) unless stated otherwise.

Characteristics	Cases (n=376)	Controls (n=6699)	Crude rate ratio (95% CI)
Mean (SD) age (years)*	68.9 (9.3)	68.9 (6.9)	—
Men*	306 (81.4)	5400 (81.4)	—
Mean (SD) duration of follow-up (years)*	4.8 (2.2)	4.8 (2.7)	—
Excessive alcohol use†	23 (6.1)	430 (6.4)	0.95 (0.63 to 1.45)
Body mass index (kg/m ²):			
<30	222 (59.0)	4171 (62.3)	1.00 (reference)
≥30	140 (37.2)	2743 (41.0)	1.08 (0.88 to 1.34)
Unknown	14 (3.7)	112 (1.7)	2.20 (1.28 to 3.78)
Smoking status:	102 (27.1)	2743 (41.0)	1.02 (1.02 to 2.22)

THIAZOLIDINEDIONES AND RISK OF BLADDER CANCER

Exclusive ever use of
pioglitazone

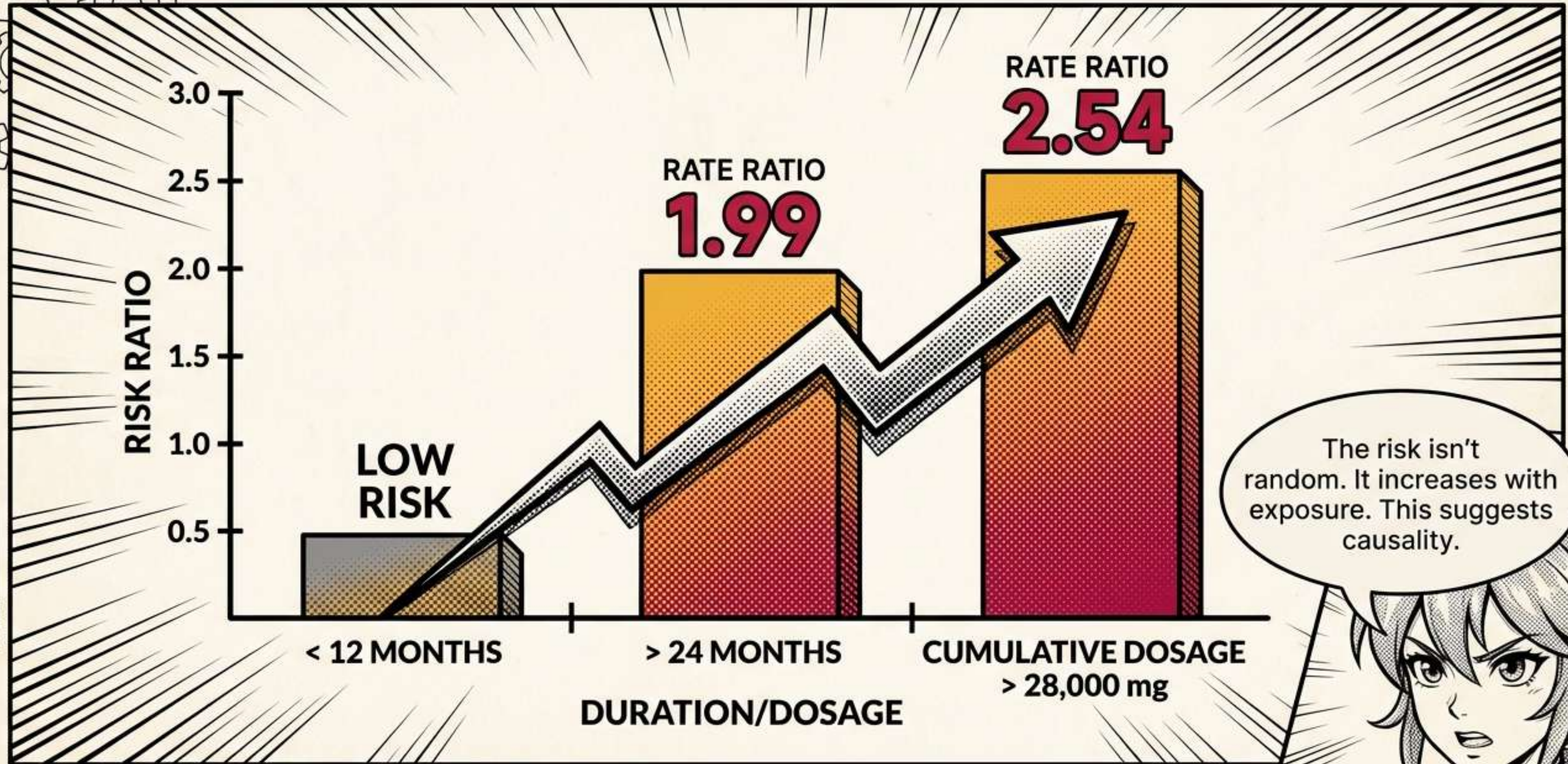
Adjusted rate ratio (95% CI)

1.83 (1.10 to 3.05)

Interpretation: Exclusive use of Pioglitazone is associated with an **83% increased rate of bladder cancer.**

Aisha:
The lower limit is 1.10.
It is statistically significant.

THE SMOKING GUN: DOSE-RESPONSE



The risk isn't random. It increases with exposure. This suggests causality.

CALCULATING THE NNH

Data table for NNH

	case	control
Stroke	74	1000
Seizure	74	5
Pain	16	
Nausea	33	
Tired	108	

$$NNH = \frac{1}{(\text{Absolute Rate Difference})}$$

$$\text{Absolute Rate Difference} = 0.022 \text{ (approx)}$$

$$NNH = \frac{1}{0.022} = \underline{\underline{4.54}}$$

Or consider the
absolute rate
difference:
74 per 100,000
person-years.

REAL WORLD CONTEXT

Salary:
4,000 Baht/month.

Current Meds:
Affordable & Effective.

Family:
4 children.

Alternatives:
Too expensive?

The cancer risk is real, but the risk of uncontrolled diabetes from stopping affordable meds might be worse for his life.

DECISION POINT

**STOP
EXPOSURE**

**CONTINUE WITH
MONITORING**

Switch to other
anti-diabetics
(if affordable).

If HbA1c benefit
outweighs Bladder
Cancer risk.

*We can stop this exposure...
and wait for further studies.*



*to take bleed
when he =*

*Location feature
no 14 km 2*



SHARED DECISION MAKING



The drug does increase the risk slightly. For every 4-5 people treated like this, we might see a problem. But your diabetes is also dangerous.

So we check the other options first?

SHARED DECISION MAKING: NNT and NNH are tools for conversation, not just calculation.

THE DOJO MASTER'S CHEAT SHEET

1. ARR (Absolute Risk Reduction) = Control Rate - Treatment Rate.

2. **NNT** (Benefit) = $1 / \text{ARR}$. (Ideal is 1).  

3. **NNH** (Harm) = $1 / \text{ARI}$. (Higher is usually safer).  

4. **LHH** (Likelihood to be Helped or Harmed) = NNH / NNT . 

5. FINAL LESSON:
Always ask 

“Compared to what?”
and “Over what
time period?” 

